due to the differential production of vacuolar anthocyanins.

In maize plant genome has Ac / Ds transposon (Ac = Activator, Ds = Dissociation). The activity of Ac element is very distinct in maize plant. The transposition in somatic cells results in the changes in gene expression such as variegated pigmentation in maize kernels. Maize genome has transposable elements which regulated the different colour pattern of kernels.

McClintock's findings concluded that Ds and Ac genes were mobile controlling elements. We now call it as transposable elements, a term coined by maize geneticist, Alexander Brink. McClintock gave the first direct experimental evidence that genomes are not static but are highly plastic entities.

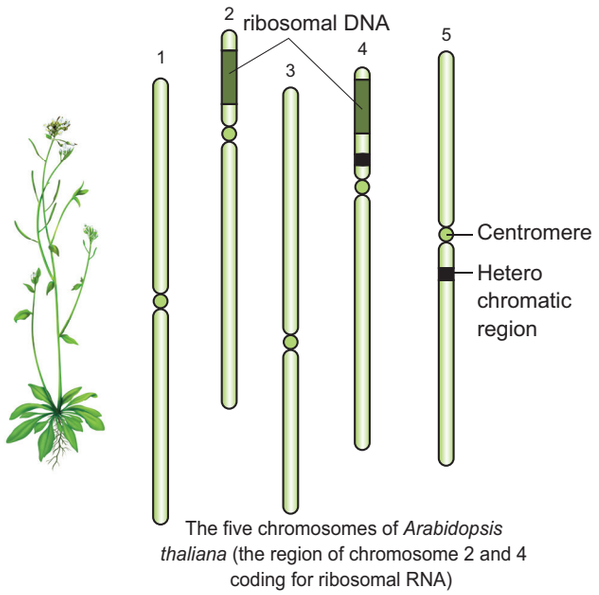
**Significance of transposons**

1. They contribute to many visible mutations and mutation rate in an Organism.
2. In evolution, they contribute to genetic diversity.
3. In genetic research transposons are valuable tools which are used as mutagens, as cloning tags, vehicles for inserting foreign DNA into model organism.

**Plant genome –** The word genome is defined as the full complement of DNA (including all the genes and the intergenic regions) present in an organism It specifies the entire biological information of an organism. There are three distinct genomes in eukaryotic cells and they are (1) The nuclear genome (2) The mitochondrial genome and (3) The chloroplast genome present only in plants.

*Arabidopsis thaliana* – Thale cress, Mouse-ear cress

1. It is a model plant for the study of genetic and molecular aspects of plant development.




The five chromosomes of *Arabidopsis thaliana* (the region of chromosome 2 and 4 coding for ribosomal RNA)

**Figure: 3.27** *Arabidopsis thaliana*

2. It belongs to mustard family and it is the first flowering plant, where its entire genome is sequenced.
3. The two regions of the nucleolar organiser ribosomal DNA which codes for the ribosomal RNA are present at the extremity of chromosomes 2 and 4
4. It is Diploid plant having small genome with 2n = 10 chromosomes. Several generations can be produced in one year. So it facilitates rapid genetic analysis. The genome has low repetitive DNA, over 60% of the nuclear DNA have protein coding functions.
5. The plant is small, self fertilizes, annual long-day plant with short-life cycle (only 6 weeks), large numbers of seeds are produced and they are easy to be grown in laboratory. It is easy to induce mutations. It has many genomic resources and the transformation can be done easily.
6. In 1982, Arabidopsis has successfully completed its life cycle in Microgravity i.e. space. This shows that Human Space Missions with plant companions may be possible.



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